

Name: _____

Date: _____

Class: _____

MyDesign Portfolio

Element D Initial Template

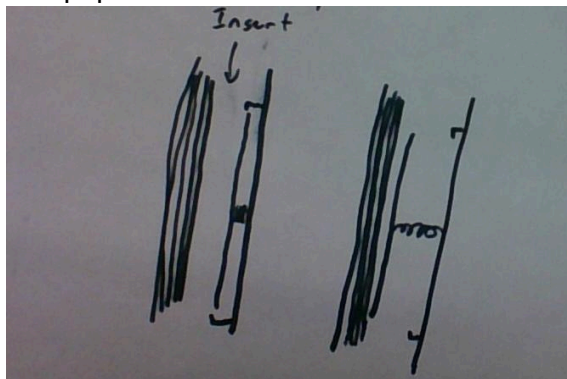
Element D: Design Concept Generation, Analysis, and Selection

Design Idea Generation (D1)

Design Idea #1

Date: 3/3/26

Description: Spring to push papers out of the way, latch to hold papers in place to make way for new paper.



Top and side view of the folder spring, folder dimensions are 12x9x1 inches. The front view is just a normal folder rectangle. Entirely made of plastic except for steel springs

Design Idea #2

Date: 3/3/26

Description: More flaps were needed for the greatest amount of protection from the destruction

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of the paper

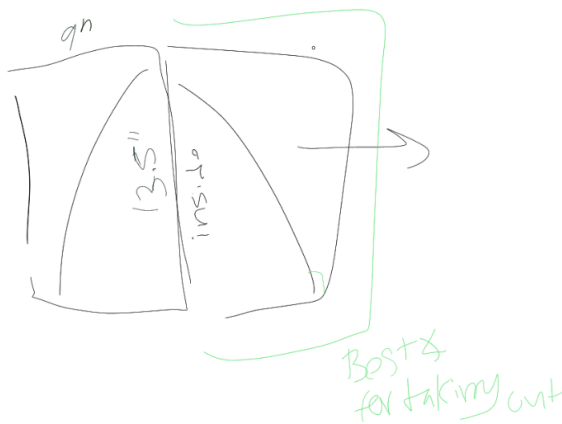


It seems that we have the same folder, but different openings in the folders, and different dimensions that indicate more flaps. More like this design, but way different.

Design Idea #3

Date: 3/3/2026

Description: A folder with the inside flaps that hold the paper designed to optimize the range of movement.



Design Idea #4

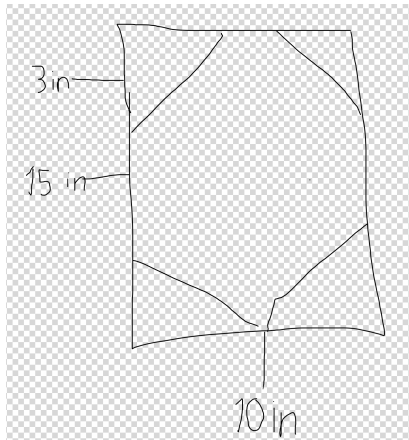
Date: 3/4/2026

Description: Flaps that hold all 4 corners of the papers, helping it stay in place while protecting it from damage. The total binder is larger than a typical sheet of paper, allowing it to fit multiple sizes of papers.

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Date: _____

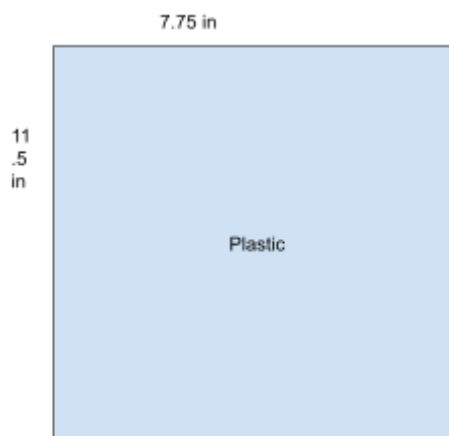
Class: _____



Design Idea #5

Date: 3/3/2026

Description: A design idea that has a slide so an individual can take it out quickly, it is slightly larger so the user can have a slightly larger margin of error when putting in paper.



Design Idea Comparison and Selection (D1, D2)

	Weight	Spring Latch	Multiple Flaps	Different Inside Flaps	Flaps that hold all four

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		Mechanism			corners of paper
Folder is not bulky	1.0 Important for convenience, but not main focus	No	Yes	Yes	Yes
Folder does not damage contents, resolves the issue of existing papers blocking new papers	1.5 Main focus of project, prevent paper damage during quick insertion	Yes	Yes	No	No
Folder can accommodate enough papers	1.0 Important for folders to continue to fulfill their main purpose	Yes	Yes	Yes	No
Folder supports quick insertion	1.25 One of the main focuses of the project, convenience of quick insertion	Yes	No	Yes	No
Folder has low friction, requires low insertion force	1.25 Supports quick insertion	Yes	Yes	Yes	No
score		5.0	4.75	4.5	1.0

Name: _____

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Class: _____

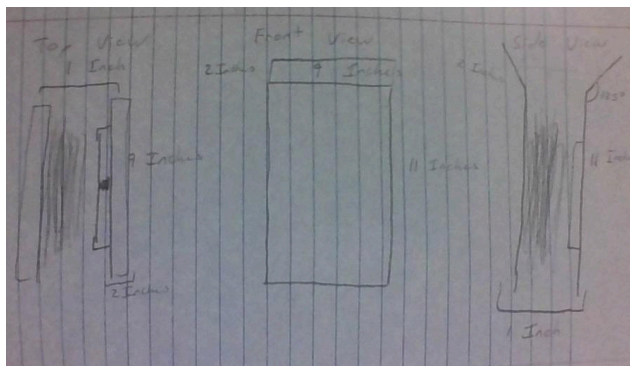
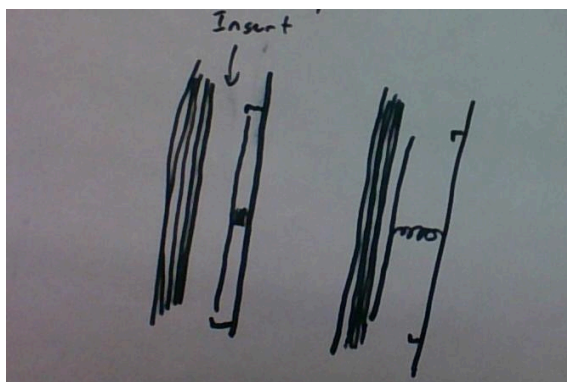
Our team chooses the design Spring Latch to solve the community problem.

Design Blueprint (D4)

Design Blueprint #1

Date: 3/3/2026

Sketch:



Top and side view of the folder spring, folder dimensions are 12x9x1 inches. The front view is just a normal folder rectangle. Entirely made of plastic except for steel springs.

Manufacturing Methods:

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Step 1: Take an existing plastic folder for the base

Step 2: Create components for the design (Insertion ramp, spring latch mechanism)

Step 3: Glue to components onto the folder

Supply List and Potential Costs:

Item	Cost each	# needed	Total
ABS Plastic filament	10 dollars	Enough for the construction of the components	10 dollars
Super glue and spring with a radius of $\frac{1}{4}$ inch, length of one inch when not compressed	For glue, 7 dollars, and for springs, 5 dollars	2-3 springs and 1 cans of super glue	17-22 dollars
Folder to act as base, standard dimensions of LxWxH 9x0.5x12 inches	5 dollars	1 folder	5 dollars
Total Cost			32-37 dollars

Plan of Action (D3)

A Gantt chart is encouraged to lay out the primary steps expected on a visual timeline. Larger tasks from the Gantt chart should be broken down further into detailed sub-tasks, which may be housed in a different day-to-day document. Discrete tasks should include enough detail such that others could replicate the efforts, such as procedures, materials, tools, status, etc. You may use the Google Sheets document at [this link](#) to start your Plan of Action.



Name: _____

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Iterations for the Solution:

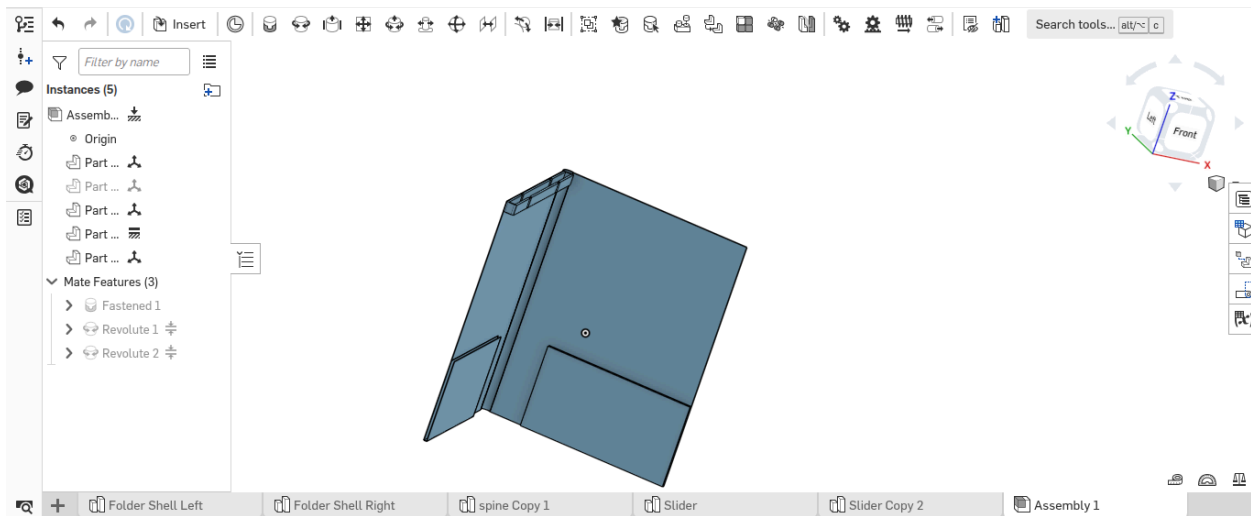
How have you changed your prototype? Record each iteration here. You may copy this blueprint for additional iterations.

Design Blueprint #2

A magnetic clip at the top of the folder to hold existing folders on one side

Date: March 23

Sketch:



Manufacturing Methods:

Step 1: Take an existing folder

Step 2: Glue the magnet clip at the top of the folder

Step 3: Glue the magnets onto the magnet clip

Supply List and Potential Costs:

Item	Cost each	# needed	Total
Small Magnets			
Folder to act as base, standard dimensions of LxWxH 9x0.5x12 inches	5 dollars	1 folder	5 dollars

Name: _____

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ABS Plastic Filament			
Total Cost			

Rubric

Element D: Design Concept Generation, Analysis and Selection						
	5	4	3	2	1	0
1. Process for Generating & Comparing Possible Solutions. The Process for Generating Possible Solutions _____.	was <i>comprehensive,</i> <i>iterative,</i> <i>and</i> <i>consistently</i> defensible (viable design highly likely)	was <i>thorough,</i> <i>iterative,</i> <i>and</i> generally defensible (viable design likely)	was <i>adequate,</i> <i>generally</i> iterative and defensible (<i>viable design</i> <i>is possible</i>)	was <i>partial or</i> <i>overly general</i> <i>and only</i> <i>somewhat</i> iterative and/or defensible (raising issues with the design viability)	was <i>incomplete</i> <i>or minimally</i> <i>iterative</i> <i>and/or</i> <i>defensible</i> (<i>raising</i> <i>issues with</i> <i>the design</i> <i>viability</i>)	<i>was not evident</i> <i>that there was</i> <i>an attempt</i>
2.Design Solution Choice. The Design Solution chosen _____.	was <i>well-justified</i> and demonstrated attention to <i>all</i> design requirements	was <i>justified</i> and demonstrated attention to <i>most if</i> <i>not all</i> design requirements	was <i>explained</i> <i>with</i> <i>reference to</i> <i>at least some</i> design requirements	was not <i>sufficiently</i> <i>explained with</i> <i>reference to</i> design requirements	<i>lacked</i> <i>support</i> <i>related to</i> design requirements	<i>was not</i> <i>attempted or</i> <i>evidence not</i> <i>provided</i>



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3. Plan of Action. The plan of action _____	<i>has considerable merit and would easily support repetition and testing for effectiveness by others</i>	<i>would support repetition and testing for effectiveness by others</i>	<i>might not clearly or fully support repetition and testing for effectiveness by others</i>	<i>has insufficient detail to allow for testing for replication of results</i>	<i>is lacking to the point that the design solution cannot be tested</i>	<i>cannot be executed or does not exist</i>
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